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1 Results

The analytical corpus comprises 6,833 emails from the inbox of a central CITF Databank coordinator, spanning correspondence with 100 participating studies from February 2021 to September 2025, involving 491 unique senders. NMF identified 14 topics organized into four thematic categories. The 14 topics identified by the model and their five most characteristic terms are shown in Figure 1. The x-axis reflects each term's NMF component weight, a non-negative score indicating the strength of association between that term and the topic, derived from the factorization of the Term Frequency-Inverse Document Frequency (TF-IDF) matrix. Higher scores indicate terms that are more strongly and distinctively associated with that topic relative to others in the vocabulary.

Figure 1: Top 5 characteristic terms and topic coherence scores (c_v) for each of the 14 NMF topics.

The topic labels, thematic groupings, and descriptions generated through LLM-assisted annotation are shown in Figure 2.

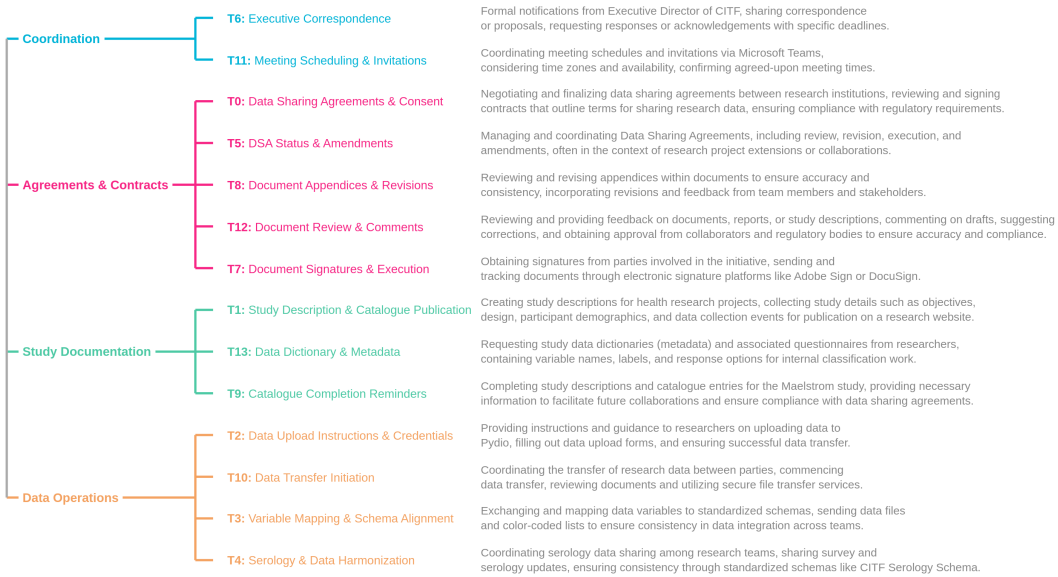

Figure 2: LLM-assisted labels, thematic groupings, and descriptions for the 14 NMF topics.

1.1 Co-occurrence Analysis of Topic Relationships

Topic co-occurrence analysis revealed two largely separate clusters of topics within the corpus, with Agreements and Contracts (5 topics, 3,793 multi-topic email assignments), Data Operations (4 topics, 2,772), Study Documentation (3 topics, 2,602), and Coordination (2 topics, 1,612) (see Figure 3). Core governance topics, including Document Signatures, Document Appendices, and DSA Status, co-occurred frequently with each other (co-occurrence scores: 0.20–0.28) but rarely with core technical topics such as Data Upload Instructions, Variable Mapping, and Data Transfer Initiation (co-occurrence scores: 0.02–0.06) This structural separation suggests that governance and data operations workstreams ran largely in parallel rather than in close coordination.

The primary point of convergence between these two clusters was Document Review and Comments, the most frequently assigned topic in the corpus (1,323 email assignments) and the most broadly connected across all thematic categories (co-occurrence scores: 0.08–0.25). This topic likely reflects the reconciliation of technical data descriptions and management requirements with institutional expectations, as data sharing agreements needed to stipulate the exact data being shared and the conditions under which they would be managed. It represents the interface between governance and operations: the point at which these concerns were most likely to be addressed together.

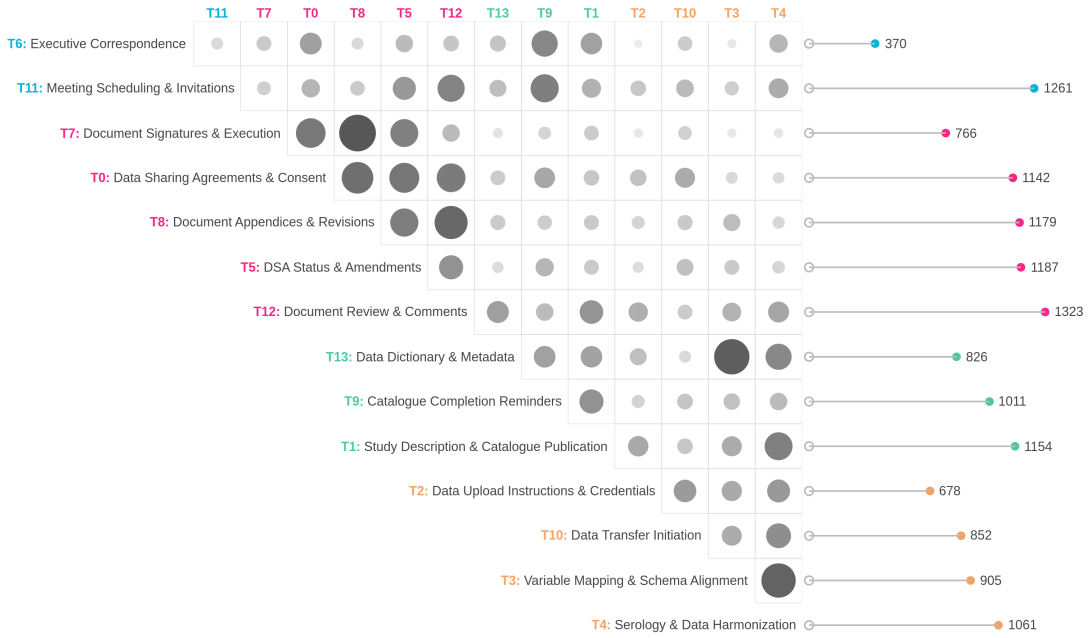

Figure 3: Topic co-occurrence matrix (0–0.30) with associated number of emails via multi-topic assignment.

1.2 Topic Distribution Across Lifecycle Phases and Sender Roles

This analysis is based on 90.3% of the full corpus (6,167 emails), with 7 studies excluded due to non-sequential milestone dates. Topic theme assignments shifted clearly across phases of the data sharing lifecycle, with Agreements & Contracts concerns proving more persistent than the phase structure alone would suggest (see Figure 4). In early phases, correspondence was dominated by Agreements & Contracts work: this theme accounted for 45% of topic assignments in the Funding Start phase and 50% in the Data Introduction phase. In later phases, Data Operations came to the fore: this theme accounted for 46% of assignments in the DSA Execution phase, rising to 56% in the Data Upload phase and 52% in the Data Cataloguing phase.

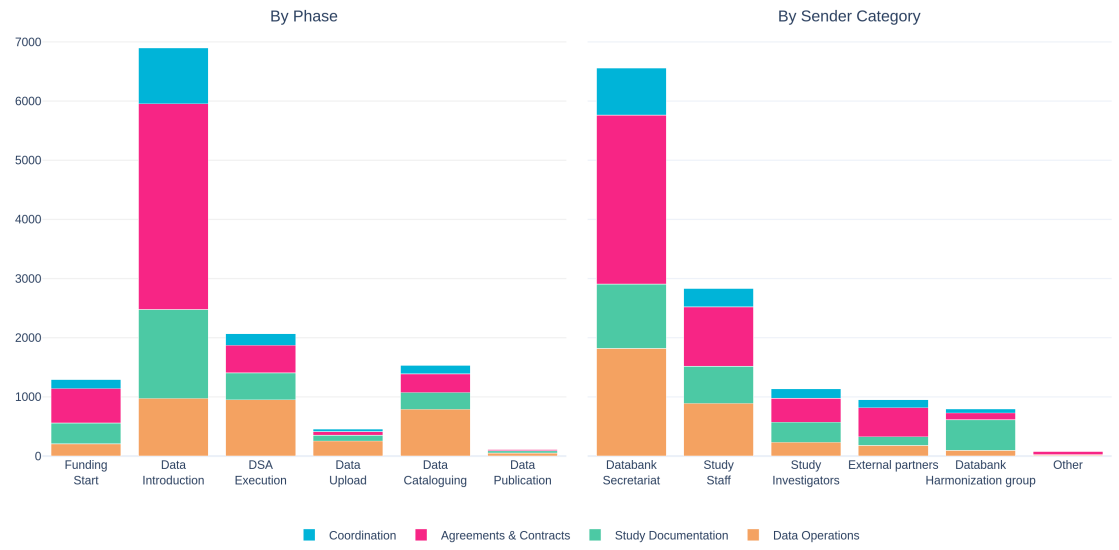

Figure 4: Topic theme assignments by lifecycle phase and sender category.

However, Agreements & Contracts concerns did not recede as the lifecycle progressed. Even in the Data Upload and Data Cataloguing phases, which were dominated by Data Operations, Agreements & Contracts still accounted for 14% and 21% of topic assignments respectively. Rather than being treated as an initial phase with a definite endpoint, contractual and administrative concerns persisted throughout the data sharing lifecycle, requiring ongoing attention well beyond the execution of formal agreements.

Studies spent a median of 395 days in DSA negotiation (Intro-to-DSA phase) and 398 days in data preparation (DSA-to-Upload phase), compared to only 22 days for technical cataloguing (Upload-to-Catalogue phase), an 18:1 ratio of administrative to technical time.

Sender analysis revealed sharply uneven operational engagement across institutional roles. The Harmonization Group sent emails at a ratio of 116 emails per sender, and the Secretariat at

101 emails per sender, compared to 7 emails per sender among External Partners, 6 among Study Staff, and 5 among Study Investigators. This concentration of email activity within Databank teams makes visible the coordination labour sustained by a small number of highly active individuals.

The topical focus of each sender group also differed markedly (see Figure 4). The Harmonization Group was the most topically specialized, with Study Documentation accounting for 65% of its topic assignments, reflecting its focused role in data dictionary and catalogue work. The Secretariat showed the most evenly distributed profile across all four themes, reflecting its broad coordinating mandate across all phases and workstreams. External Partners were most concentrated in Agreements & Contracts (52%), suggesting their engagement was primarily oriented around contractual obligations. Study Investigators showed a relatively even split between Agreements & Contracts (36%) and Study Documentation (30%), suggesting they were engaged across both contractual and documentation workstreams rather than technical data operations.

1.3 Gender Distribution of Coordination Labour

Gender analysis is based on 95.2% of the total sample (6,504 emails), with senders of unknown gender excluded. Women's representation varied substantially across topics, ranging from 36% in Serology & Data Harmonization to 70% in Study Description & Catalogue Publication (see Figure 5). Data Operations topics, including Serology & Data Harmonization (36% women), Data Transfer Initiation (40% women), and Data Upload Instructions (44% women), were the most male-dominated. In contrast, Study Documentation topics showed the highest representation of women: Study Description & Catalogue Publication (70% women), Catalogue Completion Reminders (68% women), and Data Dictionary & Metadata (65% women). Agreements & Contracts topics showed a more mixed pattern, with women accounting for 50–58% of assignments across that theme. TThis progression, from male-dominated Data Operations topics to female-dominated Study Documentation topics, is consistent with prior research documenting the feminization of invisible coordination labour in research data infrastructure [@choroszewicz2022]."

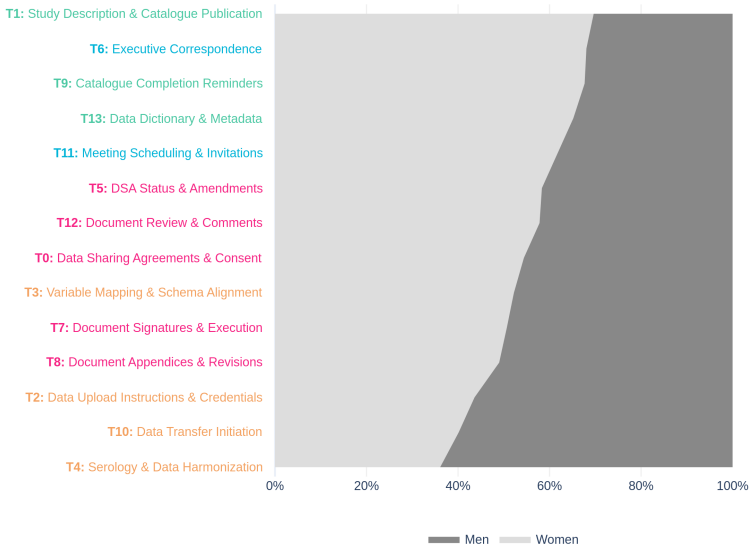

Figure 5: Topic assignments by gender category.